AnyCam 4.0.0 Audit Report

Release audit - 25 October 2026 - C32 and B75 - tag 4.0.0

A major version on CrystalHeeler's order of 2026-10-25 (an unattended run: 3.9.0, 4.0.0, 4.1.0, then push and publish 4.1.0). C32 follows CrystalHeeler's decisions of 2026-10-24, without the classifier. The object engine is a per-camera choice; the classic engine stays the default, so an update changes nothing until a camera is switched. Design and evidence: docs/C32_Object_Engine.md.

The object engine

Part	4.0.0
Objects	Changed cells up to 2 cells apart form one object (anycam_objects.find_objects): size, box, feet (bottom middle), brighter share, fill.
Where	The zone at the feet, else the zone holding most cells, else outside; Off zones and, in zones-only mode, outside do not count.
Size	obj_min_pct: 10% of the picture at sensitivity 1 to 0.05% at 100 (0.02% with the night boost), per zone.
Not bodies	Lit surfaces (one-sided, +25 grey levels, fill 0.35 or less); at night insects (one-sided, fill 0.55 or more, under 1% of the picture).
Tracking	Confirmed after 3 pictures and 3% of the width of travel; unconfirmed tracks dropped after 1 s unseen, confirmed after 3 s (B75).
Lights	Tracks dropped at a light change, a colour switch and during a pause; a 3 s light hold (classic 2 s). The 1.25 s look before recording (B85) applies.
Settings	MOTION_DEFAULTS engine = classic; validated; kept per camera in global mode (an engine-only POST); in settings copies and restores.
Answer 14	_motion_detector_kind: live, snapshots or None; the Cog panel note and a warning sign on the record button.

Evidence: the replay

409 recordings through the add-on's _motion_feed with each camera's zones and sensitivity (settings copy of 24 Oct 22:42); classic = 3.9.0.

Recordings	Classic	Objects
Headlights (29)	0	2
Insects (10)	10	8
Motion-sensitive light (8)	0	0
Dawn switch (2)	0	0
A person (6)	6	6
The cat (1)	1	1

Recordings	Classic	Objects
By day, unlabelled (228)	211	217
At night, unlabelled (122)	20	21

The 15 unlabelled files where the engines disagree were viewed: classic only, 2 insects and 2 with nothing visible; objects only, 3 people far away that the classic engine missed, 2 headlights and 6 with nothing visible.

Tests run

Suite	Result	Scope
Release gate	**PASS**	Eight gates; contracts for _motion_objects, _motion_engine and _motion_obj_place; the new file in anycam_modules.py and the Dockerfile.
Python integration	**PASS**	831 of 831; new section AW (19 checks); L1 updated for the engine setting.
JavaScript behaviour	**PASS**	221 of 221; 4 new checks for the Cog panel and the warning.
Recording replay	**PASS**	409 recordings on both engines.
Field	**NOTED**	Not run: built unattended.

Not tested

- **The object engine live.** The replay decoded files with OpenCV; its time per picture on the Pi with several cameras was not measured. On the build PC the grouping takes 0.7 ms with 1% of the picture changed, 2.4 ms at 5% and 9.7 ms at 29% (the most it groups); the Pi 4 is a few times slower, inside the 250 ms between pictures.
- **People at night far from the camera** are not in the labelled set; a white cat dashing past at night may count as an insect.
- **The Cog panel's new rows** were checked by the page tests, not on a phone screen (rule 10).

Best-practices compliance

Standard	Check applied	Result	Note
Rule 2	Logs and recordings before code	**PASS**	Tuned on 409 recordings, labelled from viewing.
Rule 7	Version and scope confirmed	**PASS**	CrystalHeeler: 4.0.0 = C32, 2026-10-25.
Imports	No cycles; new file registered	**PASS**	anycam_objects.py imports nothing from AnyCam.
P17 Testing	Tests with the change	**PASS**	Section AW; page checks.
Rule 9 (privacy)	New text	**PASS**	Made-up zone names in tests.

Files changed

File	Change
anycam_objects.py (new)	Objects, tracking, the size curve.
anycam_motion.py	The engine setting, the object path, insects, the light hold, the reduced-detection report.
anycam_zones.py	zone_map.
anycam_page.py, page_script.py	The engine select, its help, the warnings.
anycam_modules.py, Dockerfile, verify_release.py, tests/	The new file; contracts; tests.
CHANGELOG.md, config.yaml, camera_discovery.py, docs	4.0.0; the design document; the plan.